

When Stars Shine through Silence: Affective Visualization as Empowerment for Victims of Subway Harassment

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Figure 1: An overview of *When Stars Shine through Silence*, including representative visualization of (a) Stars, (b) Meteors, (c) Guiding questions, (d) Title page and (e) Policies

Abstract

Sexual harassment in public transportation remains a serious social issue, yet many visualizations focus on negative emotions, which may lead to emotional detachment. We designed *When Stars Shine through Silence*, an interactive affective visualization highlighting victims' courage and solidarity in Shanghai subway harassment incidents. Using the visual metaphor of stars, the project reveals

individual acts of resistance, supportive online responses, and evolving institutional measures, aiming to support victims toward empowerment and foster collective awareness in sexual harassment.

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1 Introduction

Sexual harassment in public transportation includes lewd comments, leering, invitations, and threats [Gekoski et al. 2017], threatening safety and inclusion. Among approaches such as public education and reporting tools, affective visualization designed to communicate and influence emotion [Lan et al. 2022], can raise public

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awareness and contributes to its prevention by creating works that resonate more emotionally. However, many existing visualizations focus on profiling perpetrators and harm [Artstein et al. 2019] which may deepen feelings of helplessness and emotional detachment. Instead, affective visualization can serve as a powerful means of *empowerment*, enabling individuals to feel capable and resourced to pursue goals [Shen and Yu 2019].

In this work, we present *When Stars Shine through Silence*, an interactive affective visualization that highlights the courage and resistance of victims in Shanghai subway sexual harassment incidents. Using the metaphor of *stars*, the work presents a starry sky showcasing the resistance behaviors of victims, the supportive atmosphere of online platforms and policy protections. This warm, affirmative approach aims to break victim stereotypes, evoke courage, and promote proactive responses and collective awareness.

2 Our Approach

2.1 Data Processing

Data Collection. We scraped posts using keywords such as “Shanghai subway sexual harassment” from Xiaohongshu, a popular Chinese social media platform with strong community engagement, retrieving metadata including *post time, title, content, number of comments, and likes*. Posts with more than 50 comments were marked as key posts with their comments further collected. We also retrieved related criminal case records from a legal database [Supreme People’s Court of the People’s Republic of China 2025] and policies issued in Shanghai since 2019—a critical temporal milestone marking the first criminal conviction for subway harassment.

Data Filtering. We collected 1,386 related posts, manually filtered those explicitly occurred in the Shanghai subway, resulting in 637 valid posts, all published after 2019. Among them, 10 were key posts with more than 50 comments, and 1,381 comments were collected. Additionally, 12 related policy documents were gathered.

Data Analysis. Focusing on the victims’ responses, we categorized five types of resistance behavior from posts: *glaring* (eye contact/staring), *evidence collection* (photo/audio/video), *physical retaliation* (grabbing/hitting), *help-seeking* (passenger/staff /police), and *none* (enduring/avoidance). Details be seen in Figure 2.

2.2 Visualization Design

We adopted *stars* as a visual metaphor (Figure 2) to represent individuals who speak out online. Like faint starlight, though each voice may appear limited in strength, collectively they construct visibility in public space, symbolizing growing courage and solidarity.

2.2.1 From Assumption to Awareness: Disrupting Common Perceptions. Before presenting the data, we introduced three guiding questions designed to provoke reflection and disrupt common assumptions about subway sexual harassment. Through Q&A mechanism, readers are encouraged to anticipate key statistics before the actual data is revealed. This approach enhances engagement and primes users for deeper interpretation of narratives that follow. Our first question, for instance, asks for the year of Shanghai’s first subway harassment conviction (Figure 1-c). The answer is 2019—unexpectedly late, given the subway has operated since 1993.

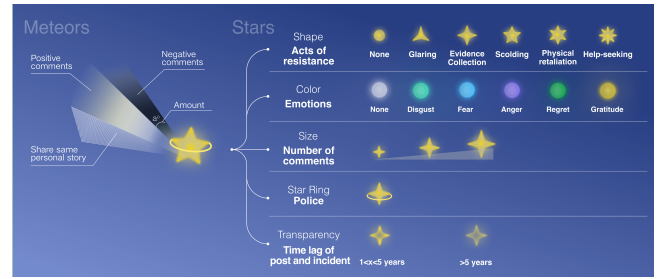


Figure 2: The visual encoding of a star and meteor

2.2.2 From Silence to Resistance: Visualizing Individual Courage. “Stars” (Figure 1-a) shows readers multi-level data of posts, including acts of resistance, emotions, and police actions, aiming to show victims’ courage. Readers are led through the visual design step by step, gaining a clear view of the data. Key cases are marked, and users can click the legend to filter data, helping focus attention and increase emotional impact. For instance, among many posts with negative feelings, some express thanks to bystanders or police who gave help. A 2023 post recalling an event ten years earlier shows how victims have grown braver to speak out.

2.2.3 From Detachment to Solidarity: Mapping Collective Support. The *Meteors’* tails encode the number of comments on key posts (Figure 1-b), with interactive elements allowing users to click and view exact values, to highlight the support and solidarity of community. We categorized comment sentiment into positive and negative to enhance users’ perception of social dynamics, and emphasized “shared experiences” to show how community support encourages speaking out. For instance, one post received 383 encouraging comments. Another featured more than 30 users sharing similar experiences, demonstrating the diffusion of collective resonance.

2.2.4 From Voices to Protection: Highlighting Institutional Response. We present the policies in Shanghai since 2019 regarding subway sexual harassment (Figure 1-e), to highlight institutional response and help users connect individual experiences to broader structural changes, reinforcing awareness of systemic protections. Practical response guidelines are also provided to enhance usability and encourage public action. For example, in 2023, Shanghai Railway Transportation Procuratorate led an inter-agency meeting mandating police to retrieve surveillance footage immediately—addressing the common challenge of evidence collection for victims.

3 Summary and Future Work

We present an affective visualization that emphasizes victims’ resistance, online support, and institutional response to encourage speaking up. Given Xiaohongshu’s predominantly female user base, the dataset inevitably reflects a certain degree of idealization. Future work will include more diverse sources for broader analysis.

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